



SpeakerSleuth: Can Large Audio-Language Models Judge Speaker Consistency across Multi-turn Dialogues?

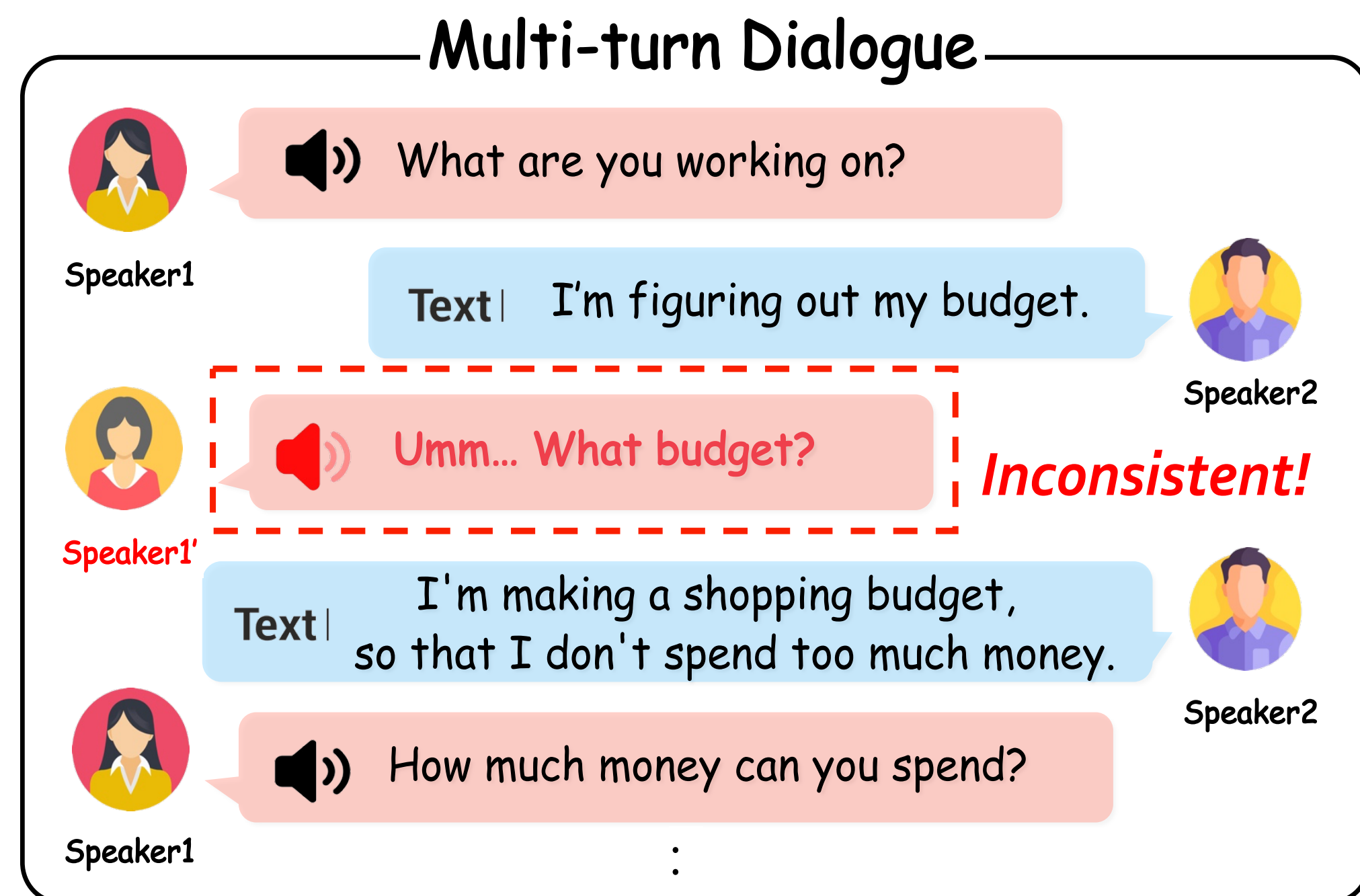
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Motivation

- Modern speech systems (e.g., voice assistants, podcast voice-overs) must keep a **consistent speaker identity** across a multi-turn dialogue: same timbre, pitch range, and voice quality.
- Existing checks rely on **pairwise embedding similarity** with hand-tuned thresholds — they can't judge a whole dialogue holistically.
- But even recent TTS models drift, showing *speaker confusion*, *timbre drift*, and *voice-quality shifts*. These failures **only surface across turns**; a single utterance sounds fine alone, yet is clearly off in the full dialogue.
- LALM judges** could process the full dialogue (audio + text) at once. But can they actually *hear* speaker inconsistency? We build **SpeakerSleuth** to find out.



SpeakerSleuth

Task Formulation

Decompose consistency judgment into three tasks that mirror a real quality-control workflow: **Detect** → **Localize** → **Discriminate**

Task 1: Detection

“Is this speaker consistent?”

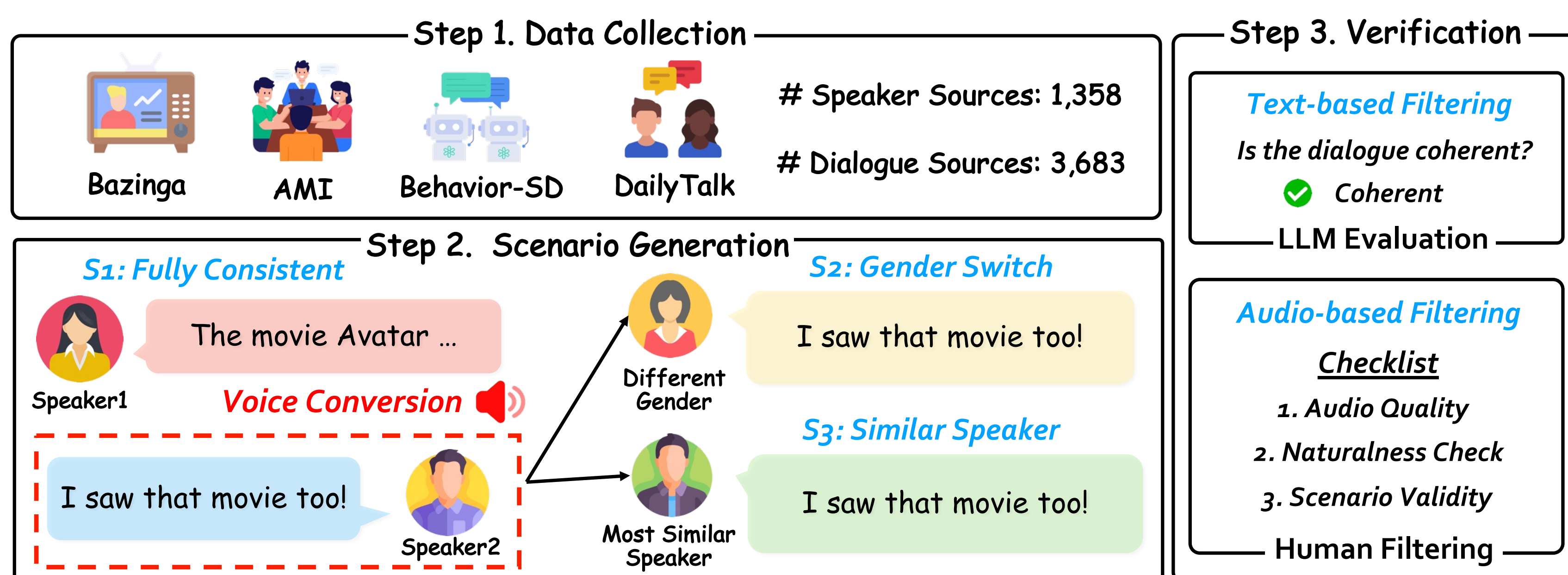
Task 2: Localization

“Which turn is inconsistent?”

Task 3: Discrimination

“Which candidate best matches the speaker?”

Benchmark Construction



Dataset	Instances	Speakers	Avg Turns [†]	Total Duration
Bazinga	636	109	9.9 ± 1.7	3.7 hrs
AMI	138	34	7.9 ± 3.4	0.9 hrs
Behavior-SD	477	52	7.9 ± 1.2	3.0 hrs
DailyTalk	567	2	9.0 ± 0.0	2.6 hrs
Total	1,818	197	8.9 ± 1.7	10.2 hrs

[†]Target speaker: 5 audio turns; other speakers: text.

Experiments

Experimental Setup

- Model:** 12 LALMs, 6 embedding methods (full results in paper)
- Evaluation Protocol:** Audio-only inputs, with a reference audio clip
- Balanced Accuracy:**

$$\text{Bal} = \frac{1}{2} \left(\text{Acc}_{S1} + \frac{\text{Acc}_{S2} + \text{Acc}_{S3}}{2} \right)$$

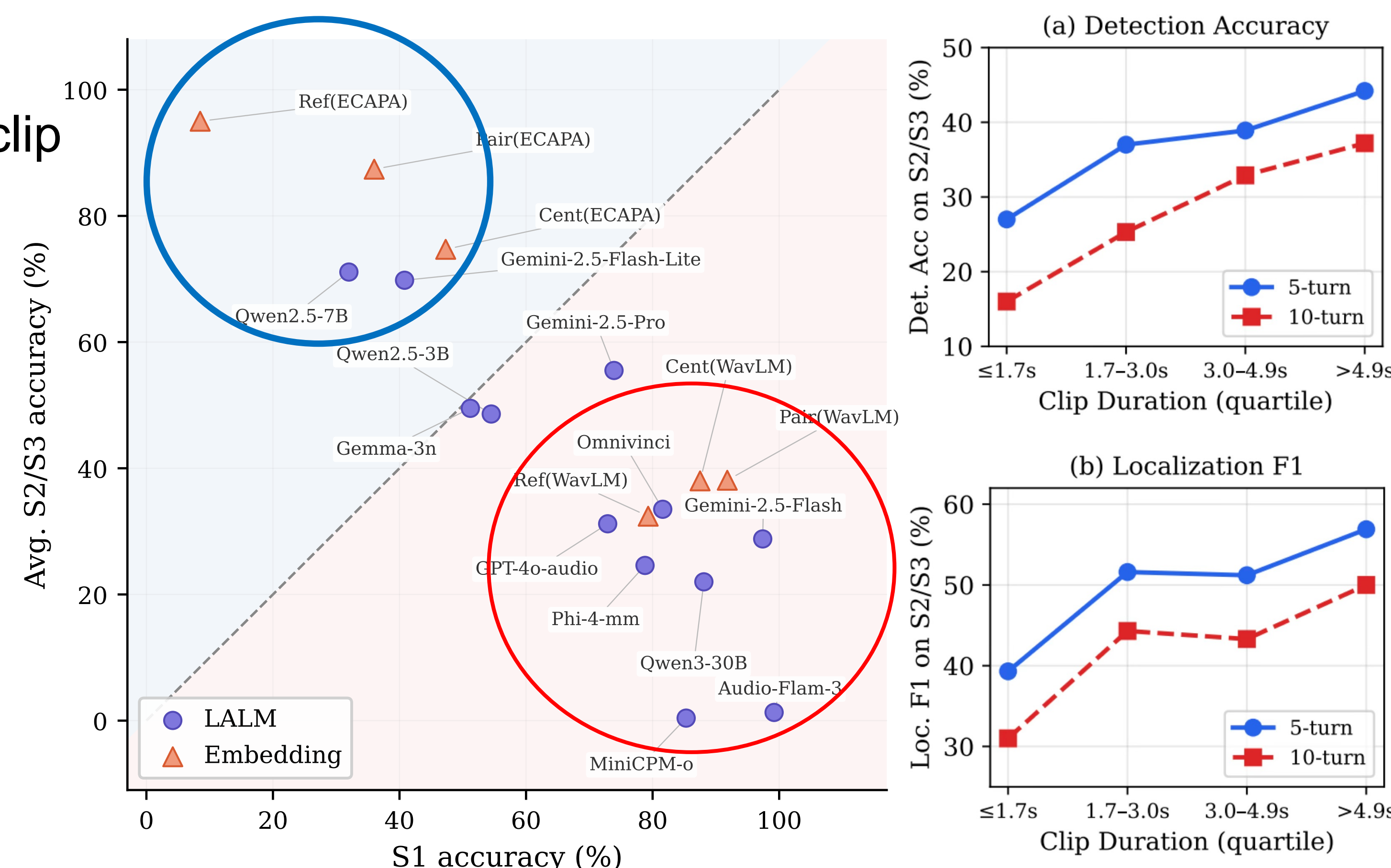
Model	Det.				Loc.		Disc.
	S1	S2	S3	Bal	Bal	Acc	
Gemini-2.5-Pro	73.9	71.6	39.3	64.7	60.1	81.5	
Gemini-2.5-Flash	97.4	45.5	12.0	<u>63.1</u>	<u>59.0</u>	<u>75.6</u>	
Qwen2.5-Omni-7B	<u>32.0</u>	72.3	69.8	51.5	29.9	42.7	
Audio-Flamingo-3	99.2	<u>1.3</u>	<u>1.2</u>	50.2	11.8	36.3	

*red/blue = over-/under-prediction

Impact of Textual Context (Localization)

- Does adding dialogue text help models focus on the target's voice?

Model	S1			S2			S3		
	Audio	+C	Δ	Audio	+C	Δ	Audio	+C	Δ
GPT-4o-audio	71.5	88.6	+17.1	15.0	3.6	-11.4	11.1	3.7	-7.4
Gemini-2.5-Pro	65.2	97.3	+32.1	62.5	7.9	-54.6	47.5	6.3	-41.2
Gemini-2.5-Flash	64.4	98.8	+34.4	62.3	30.3	-32.0	45.0	14.3	-30.7
Gemini-2.5-Flash-Lite	0.3	15.0	+14.7	36.4	29.6	-6.8	35.4	31.0	-4.4
Qwen2.5-Omni-3B	21.8	76.0	+54.2	47.7	3.7	-44.0	36.0	4.1	-31.9
Qwen2.5-Omni-7B	38.0	97.8	+59.8	22.6	0.6	-22.0	21.0	0.3	-20.7
Qwen3-Omni-30B-A3B	0.2	26.4	+26.2	23.3	36.3	+13.0	23.5	26.0	+2.5
MiniCPM-o-2.6	66.9	59.0	-7.9	35.4	23.9	-11.5	17.2	14.9	-2.3
Gemma-3n-E4B	0.0	1.3	+1.3	32.0	28.0	-4.0	32.0	27.9	-4.1
Phi-4-multimodal	99.7	100.0	+0.3	0.1	0.0	-0.1	0.1	0.0	-0.1
Omnivinci	48.3	17.8	-30.5	23.6	10.7	-12.9	15.2	9.8	-5.4
Audio-Flamingo-3	0.0	3.5	+3.5	23.3	17.6	-5.7	23.7	17.8	-5.9



Takeaway

- LALMs are not reliable judges yet**
 - Unstable internal thresholds
- Text overrides acoustics**
 - Miss even obvious gender switches when the dialogue text flows naturally
- A detection-discrimination dissociation exists**

Building reliable LALM judges requires modality balance, not just better audio encoders